

Eightfold sees the same candidate scattered across an ATS export, a recruiter's spreadsheet, a résumé, and a quick note from a phone screen. This design turns that scatter into one record per candidate — every value traceable to where it came from and how much it should be trusted. One principle shaped most of the smaller decisions: an honestly empty field is safer than a confidently wrong one.

Pipeline

Seven stages, run in order. **Confidence** is split out from **merge** deliberately — scoring how much to trust a field is a different question from deciding which value wins, and keeping them apart made both easier to test alone. **Project** runs last, so the output config can reshape results without the rest of the pipeline ever knowing it exists.

- detect** → extension + content rules sort each file into a source type
- extract** → a source-specific parser pulls fields into a raw record
- normalize** → phones, dates, skills, and country codes get standardized
- merge** → records are grouped by identity and conflicts resolved
- confidence** → each field is scored by source count and reliability
- project** → the runtime config reshapes the record into requested output
- validate** → the result is checked against the schema before returning

Canonical Schema & Normalization

Every candidate collapses to one fixed shape: identity (candidate_id , name, emails, phones, location), presentation (links, headline, years of experience), substance (skills, experience, education), and an audit trail (provenance, confidence). Four fields get standardized so the same fact always looks the same regardless of which source it came from:

phones	E.164. A bare 10-digit US-shaped number is assumed +1; anything already prefixed passes through; anything unparseable becomes null, never a guess.
dates	YYYY-MM. dateutil handles most formats; a regex fallback covers things like "Mar 2020". A year alone becomes "-01".
skills	lowercase, canonical name via an alias table (k8s → kubernetes, ml → machine learning). Unmapped skills are kept, just lowercased.
country	ISO-3166 alpha-2 via an alias table (USA / United States → US). Unrecognized values stay null rather than being guessed at.

Merge Policy & Confidence

Two records are the same candidate if they share an email, a normalized phone, or a fuzzy name match of 85% or higher, found via union-find grouping — this catches "Alice Chen" and "A. Chen" without merging two different people who share a common name. Sources are trusted in this order: **ATS JSON > CSV > Résumé / GitHub > Notes**. Scalar fields take the highest-priority value; array fields like emails and skills are unioned across every source instead of picking one winner. Every resolution, including the value that lost, is written to provenance.

A field's confidence is the share of sources that reported it, plus a 0.15 bonus when two or more agree, scaled by the most reliable source's weight (ATS 1.00, CSV 0.85, résumé 0.70, notes 0.50, GitHub 0.40). The overall score is a weighted average across fields, so three corroborating sources always outscore one low-trust note — and since the formula only looks at the inputs, the same data always scores the same.

Runtime Output Configuration

The canonical record and its output shape stay fully separate: merge and confidence never know a config exists. A projector resolves each field's from path against the canonical record — plain keys, index access (emails[0]), array projection (skills[].name), and dot-nesting (location.city) are all supported. A normalize override reformats a field afterward, and on_missing decides what happens when a field isn't there: null it, omit it, or raise.

Edge Cases, and What I Left Out

- a file is empty, unreadable, or a scanned image-only résumé logged as a warning and skipped; the run continues with the remaining sources
- a phone or date is malformed becomes null, with parse_failed recorded in provenance — nothing is invented
- two sources disagree on a value priority picks the winner; the value that lost is still recorded for audit
- the same person, slightly different name fuzzy matching (≥85%) catches it without a shared email or phone
- a record matches nobody else becomes its own standalone profile rather than being dropped

Left out deliberately: Since LinkedIn lacks a public keyless API, it was descoped. However, GitHub API lookup is fully live using public REST endpoints. Each still parses its URL and merges correctly if another source mentions the same link. Given more time, I'd add OCR for scanned résumés.